INTERNSHIP REPORT

Development of a Multiphysics Digital Twin Model of an Electric Rope Shovel for Fault Diagnosis and Prognostics

Submitted By

Ekatwa Kumar Dandsena

B.Tech Undergraduate
Department of Mechanical Engineering
National Institute of Technology Rourkela

Submitted To

[IIT KHARAGPUR]
Electrical Department

Research Supervisor:
Prof. Dr. Aurobinda Routray
Department of Electrical Engineering
Indian Institute of Technology Kharagpur

Internship Duration
May 2026 – June 2026

ACKNOWLEDGEMENT

I would like to extend my heartfelt thanks to my Professor Dr.Aurobinda Routray for giving me an opportunity to undertake this project *"Development of a Multiphysics Digital Twin Model of an Electric Rope Shovel for Fault Diagnosis and Prognostics."* I have learned a lot from him and will always remember his guidance and motivation throughout the internship period.

I would also like to thank all researchers especially my mentor Hara bhai , Sampad Bhai and other faculty members related to this project for their unwavering support and useful suggestions. Their experience has helped me in gaining more information about digital twins, mining machines, and other electromechanical systems.

I would also like to thank the Department of Electrical Engineering, Indian Institute of Technology Kharagpur for providing me with the required academic knowledge to do this research. It was due to the information obtained during academic sessions that this project was possible to complete.

Finally, I would like to thank all my family members and friends for their consistent encouragement and motivation throughout the internship period.

Ekatwa Kumar Dandsena

ABSTRACT

Electric rope shovels are among the most important and capital-intensive machines used in large-scale surface mining operations. Their productivity directly influences mine output, operational efficiency, and maintenance costs. With the growing adoption of Industry 4.0 technologies, digital twins have emerged as powerful tools for monitoring, analysis, and predictive maintenance of heavy industrial equipment.

This report presents the development of a multiphysics digital twin model of an electric rope shovel by integrating mechanical excavation dynamics with electrical drive system behavior. A geometry-based mechanical model was developed to represent bucket motion, crowd extension, hoist rope interaction, digging resistance, force decomposition, and rope elasticity during the excavation cycle. The mechanical subsystem estimates excavation forces under varying operating conditions and translates them into load demands on the hoist drive system.

The electrical subsystem models the induction motor drive responsible for hoist operation and captures the relationship between load torque, motor current, speed response, and power consumption. Coupling between the mechanical and electrical domains enables the investigation of transient loading conditions that occur during excavation.To represent realistic shovel operation, a seven-state Timed Transition Model (TTM) was developed consisting of positioning, crowding, digging, hoisting, swing loaded, dumping, and swing return states. The proposed framework captures the complete excavation cycle and enables state-dependent analysis of force distribution, payload accumulation, and drive system loading.

Simulation results demonstrate the ability of the developed framework to capture variations in hoist force, crowd force, motor torque, current demand, and power consumption throughout the digging cycle. The developed model establishes a foundation for future implementation of fault diagnosis, prognostics, and predictive maintenance strategies for electric rope shovels.

Keywords: Digital Twin, Electric Rope Shovel, Mining Machinery, Hoist System, Crowd System, Soil-Tool Interaction, Induction Motor Drive,,Timed Transition Model,Multiphysics Modeling.

Contents

1 Introduction

1.1 Surface Mining Industry

Mining is very important in the development of industries in the present age, because it supplies the raw material needed to generate power, build infrastructure, manufacture goods, transport commodities, and carry out metallurgical operations. The minerals mined include coal, iron ore, copper, bauxite, and limestone. The method of mining is determined by the geology of the mineral.

There are different ways of extracting minerals through mining processes. Open or surface mining is one of these methods which is commonly used in the process of extracting minerals that lie close to the surface of the earth. In this process, overburden, which is the layer lying above the mineral seams, is stripped off to expose the ore bed. This type of mining process is commonly used when the mineral beds are near the earth's surface.

The current modern mining activities are capital intensive and make use of machinery in large scale excavation and material handling for meeting their target requirements. With the rise in demand of minerals and energy sources, there is a need for high capacity mining machinery which can process thousands of tons of materials per hour. It is important to efficiently operate the machinery for making it cost effective and productive.

Excavation and loading are among the key processes in the mining cycle. The efficiency of this process will determine the efficiency of the whole mining process. It is for this reason that mining firms are always looking for more efficient means to monitor, automate, and control the mining processes.

In recent years, progress in sensing technologies, communication technologies, and computational technologies has allowed the application of digital technology in mining operations. Technologies like Industry 4.0, IIoT (Industrial Internet of Things), and Digital Twins are gaining popularity and becoming more relevant to increase operational efficiency, machinery performance, and predictive maintenance. All of these developments have made it possible to develop physics-based simulations of mining equipment able to mimic their operational behavior in different conditions.

Electric rope shovels are among the highly productive excavation equipment used in large-scale open-pit mines. Due to their ability to load large amounts of payload, robust design, and ability to work constantly under harsh loading conditions, electric rope shovels have become an important part of contemporary mining activities. Their knowledge about the mechanical and electrical aspects of such systems becomes a prerequisite for creating intelligent digital twins.

1.2 Excavation Equipment in Surface Mining

Mining and loading operations make up the bedrock of surface mining operations. After the overburden is stripped and the mineral resource is made available, vast amounts of material need to be mined and then loaded onto transport systems. How effectively this is done will affect mining productivity and costs.

Various pieces of excavation machinery are used in today's surface mining operations. These include hydraulic excavator, wheel loader, dragline, bucket wheel excavator, and electric rope shovel. The choice of equipment is made based on a number of different factors like deposit features, production needs, bench shape, material features, and economics, among others.

(I) Hydraulic excavators have found extensive applications because of their versatility, mobility, and selective excavation capabilities. The use of hydraulic actuators for controlling the movement of booms, arms, and buckets ensures easy operation even under diverse mining operations. However, they tend to be less productive than rope shovels of large capacities working in high production mines.

Figure 1. Hydraulic Excavator

(II) Electric rope shovels are some of the most efficient excavation equipment employed in large scale open cast mining projects. This equipment uses electric hoist and crowd systems that control the movement of the bucket during excavation. The equipment is designed in such a way that it can carry heavy loads even when it operates continuously in heavily loaded environment. Therefore, this equipment is widely used in large scale mining operations like coal, iron ore, and copper among others.

Figure 2. Electric Rope Shovel

In comparison with hydraulic excavators, electric rope shovels have much larger capacity of buckets, higher efficiency, longevity and better energy efficiency. Electric drive allows implementing more sophisticated control systems, condition monitoring and creating a digital twin. All these features make electric rope shovels suitable for intelligent mining.

Table 1 presents a general comparison between hydraulic excavators and electric rope shovels.

Table 1. Comparison of Hydraulic Excavators and Electric Rope Shovels

Parameter	Hydraulic Excavator	Electric Rope Shovel
Power Source	Diesel / Hydraulic	Electric Drive
Bucket Capacity	Moderate	Very Large
Productivity	Moderate	Very High
Operational Life	Moderate	Long
Energy Efficiency	Lower	Higher
Suitability for Digital Twin Integration	Moderate	High
Typical Application	Flexible Excavation	High-Volume Mining

Owing to their high productivity, large payload capacity, and extensive use of electrical drive systems, electric rope shovels provide an ideal platform for the development of integrated electro-mechanical digital twin models. Consequently, the present work focuses on the modeling and analysis of an electric rope shovel operating in an open-cast mining environment.

1.3 Electric Rope Shovels

One of the most massive and productive excavators that can be used in modern surface mines is the electric rope shovel. Such excavators are specially manufactured for cases when the high productivity and large capacity are necessary. This type of excavator is commonly used in the mines of coal, iron ore, copper, and other materials due to its capability to work with large amounts of materials.

Whereas hydraulic excavators use hydraulically powered hoist and crowd systems to regulate the movement of the bucket while excavating, the electric rope shovel uses electrically powered hoists and crowd systems that regulate the bucket movement. The use of electricity makes it easier to monitor and automate the excavation process due to the high torque capabilities.

A typical electric rope shovel consists of several interconnected mechanical, electrical, and control subsystems that collectively perform the excavation cycle. The major components of the machine include the boom, dipper handle, bucket, hoist mechanism, crowd mechanism, swing system, and propel system.

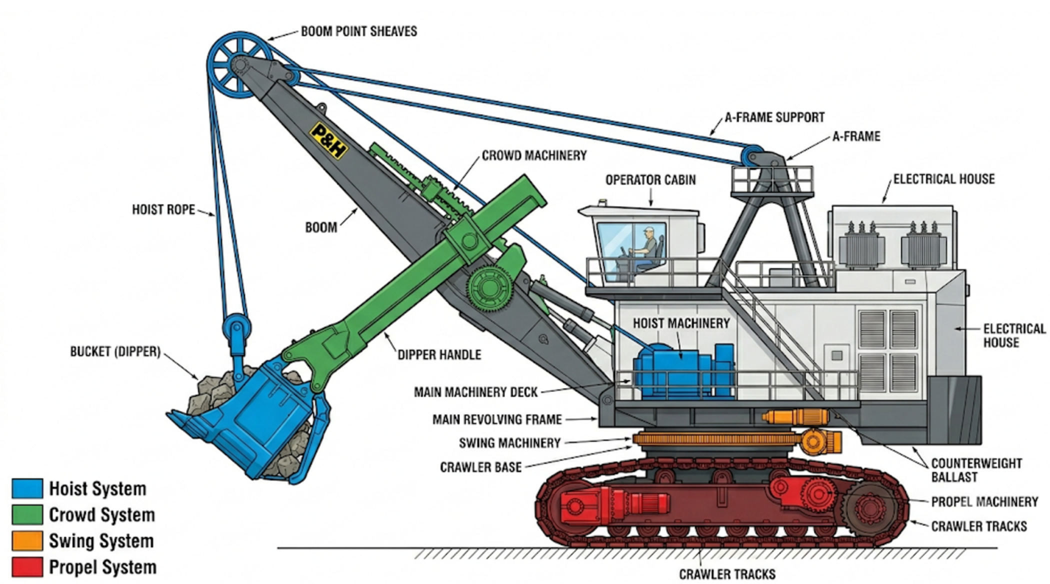


Figure 3. Shovel Architecture

The coordinated operation of these subsystems enables the execution of a complete excavation cycle comprising positioning, crowding, digging, hoisting, swing loaded, dumping, and swing return operations. Understanding the interaction between these subsystems is essential for developing a physics-based digital twin capable of reproducing the dynamic behavior of the shovel under varying operating conditions.

1.4 Major Subsystems of an Electric Rope Shovel

The electric rope shovel is a complex electro-mechanical system consisting of several interacting subsystems responsible for excavation, material handling, machine motion, and power transmission. Each subsystem performs a specific function within the excavation cycle while contributing to the overall productivity and operational reliability of the machine.

The principal subsystems considered in the present work include the boom assembly, bucket assembly, hoist system, crowd system, swing system, and propel system. Together, these subsystems enable the execution of the complete excavation cycle and form the basis of the digital twin architecture developed in this study.

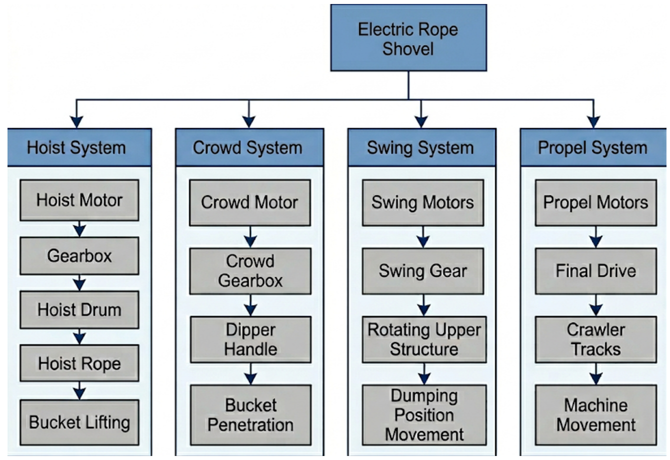


Figure 4. Shovel Subsystem

1.4.1 Boom Assembly

The boom serves as the major component for the loading of the electric rope shovel. The boom carries the hoist ropes, boom point sheaves, and other supporting structures while transferring the loads from the digging operation into the upper structure of the excavator. Due to the heavy loading conditions that occur in the process of excavation, the boom is built to be a highly durable truss structure.

The boom consists of a set of sheaves, through which the hoist ropes pass. These sheaves direct the hoist ropes and make possible the transfer of lifting forces from the hoist mechanism to the bucket. In the course of digging, significant forces that arise as a result of resistance while digging and the weight of the load are transferred via the hoist ropes to the boom.

In the developed digital twin framework, the boom point serves as an important geometric reference used for determining rope orientation, rope tension, and force transmission between the bucket and hoist system.

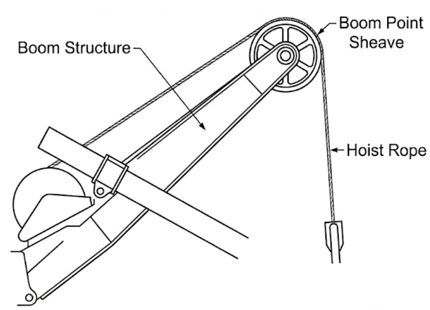


Figure 5. Boom

1.4.2 Bucket Assembly

The bucket serves as the main digging tool used by the electric rope shovel. It is a tool that is designed for penetration, cutting, and collection of materials from the excavation face. The bucket has teeth that aid in the breaking down of materials while digging.

The bucket capacity is one of the primary factors governing shovel productivity. During excavation, material progressively accumulates within the bucket, leading to an increase in payload and suspended load. Consequently, the bucket experiences excavation resistance, hoist force, crowd force, and gravitational loading simultaneously.

In this case, under the digital twin concept, the bucket is the key component used to represent the interaction between the machine and the material bank. The movement of the bucket determines the depth of excavation, digging resistance, fill factor changes, and load collection.

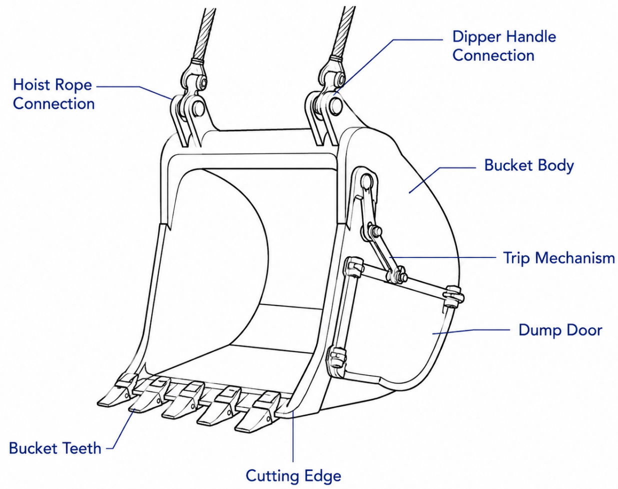


Figure 6. Bucket

1.4.3 Hoist System

The hoisting system is used to control the movement of the buckets vertically during the digging process. The hoisting system has hoisting motors, gear boxes, hoisting drums, roping systems, hoisting ropes, and boom point sheaves. Electrical energy that powers the hoisting motors is converted into mechanical torque, which is then transferred via the gear box and hoisting drums to form rope tension.

During excavation, the hoist system assists bucket motion along the digging trajectory while simultaneously supporting excavation loads. Following excavation, the hoist mechanism lifts the loaded bucket away from the excavation face and prepares it for swinging and dumping operations.

The hoist subsystem in this research is the main link between the mechanical excavation model and the electrical drive model. Hoist force calculation allows us to estimate the tension of the rope, load torque, motor current, electromagnetic torque, and energy expenditure.

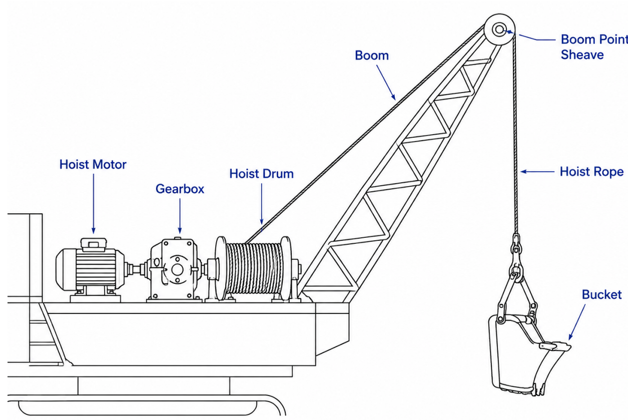


Figure 7. Hoist

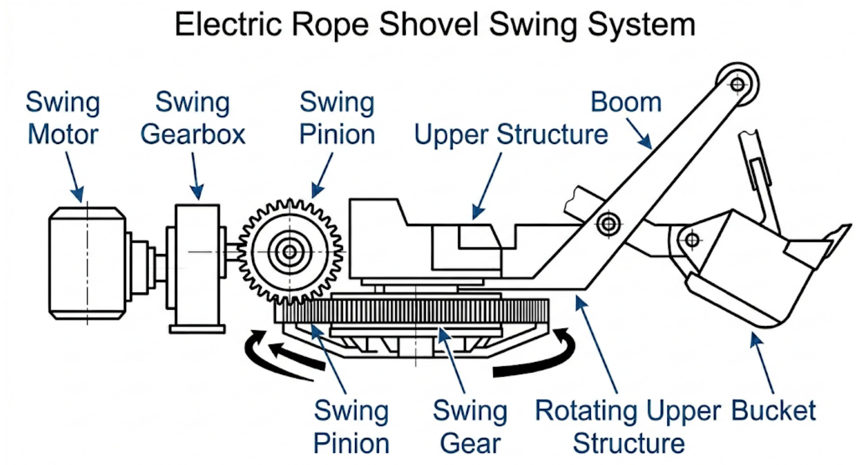


Figure 9. Swing

1.4.4 Crowd System

Control of the dipping lever for extending and retracting is achieved by the crowd mechanism. The main purpose of the crowd mechanism is to control the entry of the bucket into the material bank.

Crowd force is what provides the necessary pressure for driving the bucket into the material and maintaining the digging depth during the entire excavation process. The change in the value of the crowd force affects digging resistance, bucket position, and loading ratios of the hoist and crowd systems.

Within the developed model, crowd force is determined through geometry-based force equilibrium analysis and contributes significantly to the excavation dynamics during the digging state.

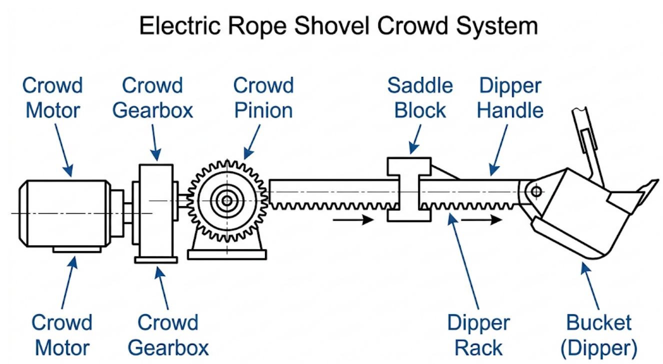


Figure 8. Crowd

1.4.5 Swing System

The swing mechanism allows for rotation of the upper part of the excavator from the point of excavation to the point of dumping. This sub-system normally comprises of the swing motors, gears, and an enormous swing bearing.

The process of swinging comes after bucket filling and lifting, whereby the excavated soil is transported to the dumping site. Once the dumping has been done, the swinging process brings back the bucket to the excavation site.

Within the Timed Transition Model, swing operations are represented by the Swing Loaded and Swing Return states.

1.4.6 Propel System

The propel subsystem enables machine movement within the mining environment. Electric propel motors drive crawler mechanisms that allow the shovel to reposition as excavation progresses.

Propel operations happen more rarely than excavation operations, but nonetheless they are necessary to facilitate access to the excavation front and sustain the mine development process as well. Since the primary emphasis in the model is on the excavation process, propel operation performance is not taken into account in the current digital twin.

2 Electrical Power Architecture of the Electric Rope Shovel

The extraordinary digging capacity of the latest generation of electric rope shovels can be attributed to their intricate electrical power system, which helps them deliver large amounts of power to several drive systems at once. Unlike the conventional excavators driven by diesel engines, electric rope shovels extract electricity from the mine power distribution system and control it using an advanced power conversion system.

The electrical power system serves as the primary source of energy for the hoist, crowd, swing, and propel mechanisms. Through a combination of transformers, power electronic converters, motor drives, and control systems, electrical energy is converted into the mechanical power required to perform excavation, lifting, swinging, and machine positioning operations.

A typical electric rope shovel electrical architecture consists of the following major subsystems:

- Incoming power supply system
- Main transformer
- Rectifier and DC link subsystem
- Inverter drive subsystem
- Motor drive architecture
- Auxiliary electrical systems

The interaction between these subsystems enables efficient power transmission from the mine grid to the excavation mechanism while providing the controllability required for modern digital mining operations.

2.1 Incoming Power Supply System

The electric rope shovels work under highly stressful power conditions, thus requiring their own power supply from the source that can provide high voltages. For most mines, electricity is supplied by the mine's electricity grid through trailing cable systems that are capable of providing several megawatts of power.

Typical rope shovels receive electrical power at medium-voltage levels ranging from 6.6 kV to 13.8 kV depending on machine size and mine infrastructure. The use of electrical power eliminates the need for large diesel engines while enabling higher efficiency, reduced operating costs, and improved environmental performance.

Incoming electrical supply acts as the main energy input to all digging and movement operations of the machine. This means that power quality and stability become very crucial for ensuring consistent shoveling activity.

2.2 Main Transformer

The main transformer performs voltage transformation and electrical isolation functions within the shovel power system. Incoming medium-voltage power is converted to voltage levels suitable for the various drive and auxiliary systems installed on the machine.

Besides the transformation of the electric current's voltage, the transformer also ensures electrical insulation between the mine electric power system and electrical equipment used on board.

The transformer acts as the first stage of power conditioning and serves as the interface between the mine electrical infrastructure and the shovel electrical architecture.

2.3 Rectifier and DC Link Subsystem

The electric energy provided from the mine distribution system is present in the form of alternating current (AC). But the advanced motor drive systems for varying speeds demand a steady DC supply. As such, the input AC voltage is first transformed to a DC voltage through rectification process.

The rectifier subsystem serves as the interface between the transformer and the motor drive system. Its primary function is to convert the transformed AC voltage into a DC voltage suitable for subsequent power electronic processing. Depending on the machine design, diode rectifiers, thyristor rectifiers, or active front-end converters may be employed.

After rectification, the energy gets stored temporarily in the DC Link sub-system. The DC link generally includes large capacitors and filters that are used to regulate voltage spikes and form an energy source for the inverter drives. This energy storage process provides decoupling between the supply end and the motor drive end.

The DC link plays a critical role in maintaining continuous operation during transient loading conditions. Rapid variations in excavation loads, hoist forces, and motor torque demands can be accommodated through the energy buffering capability of the DC link, ensuring reliable operation of the electrical drive system.

2.4 Inverter Drive Subsystem

The inverter sub-system takes care of conversion of the DC link voltage to an AC voltage that is used to drive the induction motors that form part of the bucket-wheel shovels. The speed, torque and performance of the motors are controlled through modulation of the voltage levels and frequencies.

Modern electric rope shovels utilize variable frequency drive (VFD) technology to achieve efficient motor control under varying operating conditions. The inverter employs high-speed switching devices to synthesize three-phase voltages with controllable frequency and amplitude. Through this process, the electrical drive system can adapt to changing excavation loads while maintaining desired performance characteristics.

The inverter subsystem acts as the basis for modern control methods for motors like Field Oriented Control (FOC) that facilitates separate control of flux and torque producing currents. It becomes highly significant for excavators as there are considerable variations in torque load during its operation.

The output of the inverter directly supplies the hoist, crowd, swing, and propel motors, thereby establishing the link between power electronics and mechanical motion generation.

2.5 Motor Drive Architecture

The electric rope shovel employs several electric drives to carry out the duties of digging, handling materials, rotating, and moving. Each of the subsystems has its own motor and drive electronics tailored for its unique function.

The hoist drive system represents the most critical excavation-related drive subsystem. It is responsible for generating the rope tension required to lift and transport the bucket throughout the excavation cycle. Due to the large payloads involved, the hoist motors are designed to deliver high torque and operate under severe loading conditions.

The crowd drive system controls dipper handle extension and bucket penetration into the material bank. During excavation, the crowd motor works in conjunction with the hoist system to maintain the desired digging trajectory.

The swing drive system supplies rotational motion for the upper assembly from the point of excavation to the dumping point. After the buckets are filled, the swing motors move the material to be dumped and bring the empty buckets back to the excavation point.

The propel drive subsystem enables machine repositioning within the mine. Although propel operations occur less frequently than excavation operations, they are essential for maintaining access to new excavation zones as mining progresses.

The coordinated operation of these drive systems enables the execution of the complete excavation cycle while providing the flexibility required for varying operating conditions.

2.6 Auxiliary Electrical Systems

Besides the main excavation drive units, electric rope shovels have additional electrical subsystems which help to ensure the safety and efficiency of work of the machines. The additional subsystems ensure such functions as cooling, lubricating, illumination, control and protection of the electric rope shovel from external influences.

Auxiliary transformers are employed to supply electrical power to low-voltage equipment and support systems. These include machine lighting, control electronics, operator cabins, instrumentation, communication equipment, and maintenance systems. Separate lighting transformers are often provided to ensure reliable illumination during night operations and low-visibility conditions.

The cooling system is key to ensuring that the operating temperature remains at an appropriate level for the motor, power electronics converters, transformers, and electrical panels. For that reason, specialized cooling blowers are incorporated into the electrical system design.

The use of the lubrication system helps decrease the wearing out of gears and bearings among others. Lubrication pumps which are electrically operated ensure that the lubricating oil is continually moving within the mechanical system.

Modern rope shovels also employ Human Machine Interface (HMI) systems for real-time monitoring and diagnostics. These systems provide operators and maintenance personnel with information regarding machine status, electrical parameters, fault conditions, and operational performance. Communication networks further enable data exchange between onboard controllers, sensors, and supervisory systems.

Collectively, these auxiliary systems ensure the safe and efficient operation of the shovel while supporting advanced monitoring, fault diagnosis, and digital twin implementation.

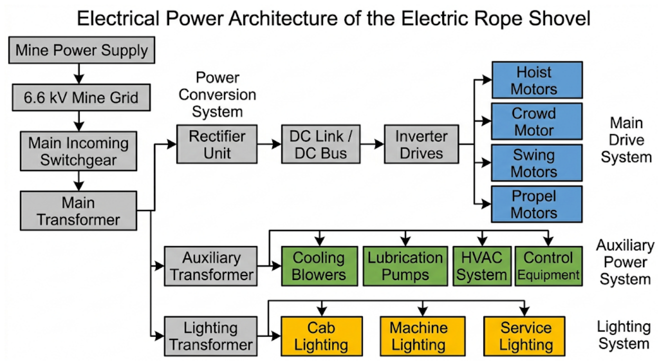


Figure 10. Overall electrical power architecture of the electric rope shovel showing the power flow from the mine grid to the excavation subsystems

3 Digital Twin Architecture of the Electric Rope Shovel

A Digital Twin is a virtual representation of a physical system that continuously reproduces its behavior through mathematical models, data integration, and computational analysis. In mining applications, Digital Twin technology enables virtual monitoring, performance assessment, fault diagnosis, and operational optimization of critical equipment.

The aim of the current research is to create a digital twin of an electric rope shovel based on physical modeling, which will be able to simulate the processes of mechanical mining and the behavior of the electrical drive. To fulfill this aim, the structure of the digital twin is created in several interacting layers of physical machine, mechanical dynamics, electrical drive, controls, and system integration.

The architecture presented makes it possible to convert the excavation load into the electrical drives response while, at the same time, tracking the operational state of the shovel during the excavation process. This hierarchical approach increases the model modularity and allows for further incorporation of sensor data into the system.

3.1 Physical Layer

The physical layer represents the actual electric rope shovel operating within the mining environment. This layer comprises the mechanical structure, electrical drive systems, excavation mechanism, sensors, and supporting auxiliary equipment.

During the working process, the shovel acts on the material bank directly using the bucket and digging mechanism. The forces created during digging are transferred through the hoist and crowd mechanisms to the drive motors. At the same time, the electric energy provided from the mine power system is converted into mechanical energy for carrying out the excavating works.

All the variables that are used in the Digital Twin come from the physical layer itself. These include the position of the bucket, digging depth, load, lifting force, motor current, torque, speed, and power. All these parameters come from the physical layer and are used to create a digital twin of the system.

The physical layer represents the actual electric rope shovel operating within the mining environment. This layer comprises the mechanical structure, electrical drive systems, excavation mechanism, sensors, and supporting auxiliary equipment.

The bucket and excavation mechanism allow direct contact between the shovel and the material bank. The forces created due to excavations are transferred via the hoist and crowd mechanism to the drive motors. Meanwhile, the energy produced in the form of electricity from the mine's power system is converted into mechanical energy.

The physical layer acts as the source for all operational parameters that are mirrored in the Digital Twin system. Parameters like the position of the bucket, dig depth, load, lifting force, motor current, torque, speed, and power are some of the operational parameters that are generated from the physical system.

3.2 Mechanical Modeling Layer

The mechanical modeling layer reproduces the excavation behavior of the shovel through a physics-based representation of bucket motion, digging resistance, payload accumulation, and force transmission.

This level integrates the bucket kinematics, creation of excavation trajectory, digging resistance calculation, fill factor variation, calculation of hoist force, crowd force, and load torque generation. This gives the developed model the dynamics of interaction between the excavation process and the material bank.

Some examples of outputs from the mechanical layer are the excavation resistance, bucket loading, hoisting effort, crowding effort, and motor loading torque. The above are inputs to the electrical drive system and form the connection between the mechanical and electrical layers.

3.3 Electrical Modeling Layer

Electrical modeling layer is the modeling of the electrical to mechanical energy transformation carried out by the shovel driving mechanism. Electrical modeling layer is the duplication of the dynamic processes in the induction motors, power electronics and control systems used to generate torque needed for excavation.

The main input for the electrical system model is the load torque provided by the mechanical modeling system. Changes in excavation resistance, load build-up, hoist force, and crowd force get converted into changes in the required motor load torque. The electrical system calculates the values for the motor current, electromagnetic torque, speed, and energy requirement.

The electrical subsystem incorporates motor shaft dynamics, rotor flux dynamics, electromagnetic torque generation, current control, speed control, and flux regulation mechanisms. Through these models, the Digital Twin is capable of reproducing the electrical response of the shovel under varying excavation conditions.

Outputs from the electrical domain include motor current, electromagnetic torque, rotational speed, electrical power, and performance parameters. This set of parameters offers information on loading and energy consumption during excavation.

3.4 Control and State Recognition Layer

The control and state recognition layer governs the operational behavior of the Digital Twin through a Timed Transition Model (TTM). This layer is responsible for identifying and reproducing the sequence of operational states encountered during a complete excavation cycle.

The proposed Timed Transition Model models the excavating process through seven different operational states which include Positioning, Crowding, Digging, Hoisting, Swing Loaded, Dumping, and Swing Return. The transitions among these states are based on various physical parameters including excavation depth, bucket fill ratio, hoist height, swing angle, and total operational time.

The state recognition framework enables the Digital Twin to reproduce realistic machine behavior by activating the appropriate mechanical and electrical models corresponding to each operational state. Consequently, the Digital Twin can capture the changing load conditions, force distributions, and motor operating characteristics that occur throughout the excavation cycle.

The incorporation of state recognition logic greatly increases the physical reality of the model and creates the basis for future implementation of automatic control, evaluation and prediction maintenance techniques.

3.5 Digital Twin Integration Layer

In the Digital Twin Integration Layer, the physical, mechanical, electrical, and control systems are integrated into one simulation environment. By means of this integration, information is being shared continuously between the different modeling layers, which allows for the replication of the entire electro-mechanical behavior of the shovel.

Within the proposed framework, excavation resistance generated by the mechanical subsystem influences the hoist and crowd forces acting on the bucket. These forces are subsequently converted into load torque acting on the electrical drive system. The electrical model responds by generating motor current, electromagnetic torque, speed, and power characteristics consistent with the imposed loading conditions.

On the other hand, the Timed Transition Model regulates the operational state transitions and ensures the proper behavior of the subsystems during the excavation process. Such an architectural combination enables the Digital Twin to model both the physical excavation process and the electrical drive system response to it.

The resulting framework provides a foundation for performance analysis, operational state recognition, fault diagnosis, predictive maintenance, and future real-time Digital Twin implementation in large-scale mining equipment.

4 Literature Review

The creation of a Digital Twin for an electric rope shovel involves the application of information coming from various fields of research such as excavation mechanics, modeling of digging resistance, electromechanical drives, operating conditions detection, and Digital Twin. Researchers have explored numerous topics related to the functioning of mining equipment over the last decades in order to increase its efficiency and reliability.

Early studies primarily focused on the geometric and kinematic analysis of excavation machinery. These investigations established the fundamental relationships governing bucket motion, excavation trajectories, and force transmission mechanisms. Subsequently, research efforts expanded toward dynamic excavation modeling, digging resistance estimation, and machine–material interaction analysis.

The availability of sensing devices and computing capacity has caused recent studies to be more inclined toward development of integrated electromechanical models that will help in developing a relationship between excavation and electrical drive dynamics. This development has made possible the development of monitoring systems, condition assessment techniques, and Digital Twin models for large mining machinery.

This section reviews the major contributions reported in the literature and identifies the research gaps that motivate the development of the proposed physics-based Digital Twin framework.

4.1 Excavation Kinematics and Digging Mechanics

Kinematics of excavation represents the movement of the bucket while interacting with the bank of materials during the excavation operation. The precise description of the bucket movement is important for prediction of excavation force, payload accumulation, excavation resistance, and machine productivity.

Early excavation models represented bucket motion using simplified geometric relationships based on hoist and crowd movements. These approaches enabled the calculation of bucket position and excavation depth but often neglected the interaction between bucket motion and excavation resistance.

Further research developed a more advanced kinematics model that would be able to account for the entire course of the excavations. It was based on the effects of hoist, crowd, and bucket positioning on the digging process. This resulted in more accurate estimations of the depth of excavation and its engagement into the material.

The mechanics of excavation involve a complex interaction between the bucket and the material bank. During digging, the bucket experiences resistance forces arising from material cutting, material lifting, friction, and inertia effects. The magnitude of these forces depends on several factors including material properties, excavation depth, bucket geometry, and excavation velocity.

Many researchers have proven that excavation productivity greatly depends on the digging path of the bucket. The right path selection can lower excavation resistance, enhance bucket filling rate, and save energy. For this reason, most recent models of excavators use path-dependent formulas to estimate the forces of excavation.

Accurate modeling of the kinematic of excavation is thus the basis for computing forces and payload as well as the digital twin development process. The current study focuses on the modeling of the bucket motion by an excavation trajectory which can help estimate the depth of excavation, fill factor evolution, and forces transfer during excavation.

4.2 Digging Resistance Modeling

The estimation of digging resistance represents one of the most important aspects of excavation modeling because it directly influences machine productivity, power consumption, structural loading, and component life. Digging resistance corresponds to the force opposing bucket motion during excavation and is generated by the interaction between the bucket and the material bank.

Many researchers have developed methods of estimation of the forces of digging process. The earliest methods considered excavating resistance as a constant load applied on the excavator bucket. Despite computational efficiency, this method could not account for changes in material characteristics and excavation conditions.

More advanced models decomposed the excavation force into multiple components associated with material cutting, material lifting, frictional effects, and inertial forces. These formulations improved the representation of soil-tool interaction and enabled more realistic predictions of machine loading conditions.

Further studies included parameters like density, cohesion, internal friction angle, and depth of cutting in the calculation of force required for excavation. It was concluded from these methods that there is a continuous variation in the resistance of excavation and depends upon geometry and operation of excavation.

However, recent research has been directed toward physics-based force models that can predict the dynamic forces acting in the process of excavation. These models yield important data on the forces transmission, loading of the bucket, loading of the drive system, and energy expenditure. Hence, calculation of digging resistance is an integral part of contemporary digging simulation.

4.3 Electrical Drive System Modeling

The performance of an electric rope shovel is strongly dependent on the behavior of its electrical drive system. Hoist, crowd, swing, and propel operations are all accomplished through electrically driven motors capable of delivering high torque under varying load conditions.

Early electrical drive models employed simplified steady-state representations that focused primarily on power consumption and average operating characteristics. While useful for preliminary analysis, such approaches were unable to capture transient responses associated with excavation loading and motor control dynamics.

The invention of power electronic converters and new methods of control gave rise to dynamic motors models that can mimic the behavior of currents, torques, speeds, and fluxes in various operating conditions. Using such models, it became possible for scientists to study the effect of mechanical load on drive performance.

Induction motors are widely used for large-scale industrial operations due to their durability and power capacity. Hence, many research works have been done on the modeling of dynamics of induction motors, methods of generating torque, flux control, and variable speed drives.

Modern electrical drive models frequently employ vector control and field-oriented control techniques to achieve independent regulation of flux-producing and torque-producing current components. Such approaches provide superior dynamic performance and enable efficient operation under rapidly changing load conditions.

Combination of models for electric drives with those for mechanical excavation is becoming increasingly crucial in the design of Digital Twins for mining machines. Integration of such models allows analysis of loads on the excavation, motor behavior, power consumption, and efficiency in one go.

4.4 Digital Twin Technology

Digital Twin technology has emerged as one of the most significant developments associated with Industry 4.0 and smart manufacturing initiatives. A Digital Twin can be described as a virtual representation of a physical asset capable of reproducing its behavior through mathematical models, sensor data, and computational analysis.

While the idea was first implemented in the aerospace and manufacturing industries, it has now been adopted in energy systems, transportation systems, industrial machines, and mining. The main purpose of Digital Twins is to increase system comprehension, boost efficiency, and aid in making predictive decisions.

The contemporary Digital Twin generally consists of physical models, data-driven approaches, sensor readings, and communication technologies to form an ever-evolving virtual image of the physical system. Depending on its usage, Digital Twins can be applied in various applications, including performance monitoring, fault detection, maintenance, optimization, and lifecycle management.

In terms of mining equipment, the Digital Twin approach gives options to increase efficiency in using machines, to lower maintenance expenses, and to avoid any unplanned downtime. In fact, with both mechanical and electrical system models, it is possible to continuously monitor the status of machines.

The increasing availability of sensing technologies, industrial communication networks, and computational resources has accelerated the adoption of Digital Twin frameworks for large-scale mining equipment. Consequently, Digital Twin technology has become a promising tool for enhancing the reliability and efficiency of modern mining operations.

4.5 Operational State Recognition and Excavation Cycle Modeling

The operational behavior of excavation machinery can be represented as a sequence of distinct states corresponding to different phases of the excavation cycle. Accurate identification of these states is important for performance assessment, productivity analysis, energy consumption estimation, and Digital Twin development.

Many scientists have suggested the application of the state-based approach to the modeling of excavation processes. The method includes the division of the excavator process cycle into several operation modes like digging, lifting, swinging, dumping, and return movements. Transition between states is made based on machine location, force measurement, timing data, etc.

State-based modeling provides several advantages over continuous representations of machine operation. By identifying the active operational state, it becomes possible to associate specific mechanical loads, force distributions, and power consumption characteristics with each phase of the excavation cycle. This improves the interpretability of simulation results and enables more realistic representation of machine behavior.

Newer literature has emphasized the significance of merging state recognition within operational systems into Digital Twin models. This is because it helps the virtual representation mimic the performance of machinery in different operating states while still maintaining alignment with the real digging process.

Regarding the application to electric rope shovels, state identification becomes especially critical since there are variations in terms of load during excavation, condition of the load, hoist forces, and operation of motors during the entire process of excavation. Hence, a state-based model becomes an efficient tool for describing the interrelation between mechanical and electrical systems.

4.6 Research Gap and Motivation

According to the literature review, considerable advancement has been made in the field of mechanics of excavation, prediction of digging resistance, electric drives, and Digital Twin. However, most of the studies conducted till now have dealt with subsystems separately rather than considering the entire process of excavation.

Many excavation models emphasize bucket kinematics, digging resistance, and force estimation without considering the electrical drive system response. Conversely, electrical drive studies frequently focus on motor dynamics, control strategies, and power electronics while neglecting the influence of excavation loads generated during digging operations.

Moreover, many implementations of Digital Twins make use of data-driven techniques but do not contain a physical model of the excavation. Consequently, the interaction between the excavation mechanics, cargo loading, force transmission, and electric drive operation cannot be captured.

Another restriction that can be seen in the literature is the lack of implementation of operational state recognition into shovel models. Due to the variation in load conditions, force distributions, motor torque needs, and energy requirements during an entire excavating process, the use of state-dependent modeling becomes crucial.

To overcome these limitations, the current research proposes the design of a Digital Twin framework that considers the mechanical model for the excavation process, an electrical model for the drive system, and operational state recognition. To model the entire excavation cycle, the Timed Transition Model that consists of seven states is used and allows analyzing the following factors such as excavation resistance, load accumulation, hoisting force, crowding force, motor torque, current consumption, and energy expenditure.

The proposed framework establishes a foundation for future implementation of advanced monitoring, fault diagnosis, prognostics, and predictive maintenance capabilities in electric rope shovels.

5 Timed Transition Model for Excavation Cycle Representation

The workings of an electric rope shovel involve a series of repeated tasks that make up the entire excavation process. In normal operations, the equipment is always in motion as it carries out excavation, lifting, material movement, dumping, and back tracking in order to reach the required production rate. Due to their dynamism, these processes lead to changes in excavation forces, payload, motor torque, and power usage.

Continuous models are not always capable of modeling such clearly distinguishable phases in a physical sense. Therefore, a model based on states was used in the current research. The framework involves a use of Timed Transition Model (TTM), whereby the entire process of excavation is defined in terms of discrete states linked via physical transition criteria.

The TTM helps in the replication of real behaviors of the shovel through the association of distinct physical and electrical properties in each operational mode. In addition, the model aids in the identification of operational modes and provides an approach for coupling excavation mechanics with electrical drives dynamics.

The entire process of excavation cycle is illustrated through seven separate states of operation which include:

Positioning, Crowding, Digging, Hoisting, Swing Loaded, Dumping and Swing Return. The seven operations are those activities that are necessary for an entire excavation cycle.

5.1 Excavation Cycle Overview

The excavation cycle begins with the positioning of the bucket near the excavation face. The crowd mechanism subsequently advances the bucket toward the material bank, enabling penetration into the material. Once contact is established, the digging phase commences and material progressively accumulates within the bucket.

Once the excavating process is completed, the hoisting system lifts the filled bucket away from the face of excavation. Then, the upper structure swings towards the dumping point in what is known as the swing-loaded process. Once the dump point is reached, the payload is dumped into it.

After dumping, the swing system returns the empty bucket to the excavation face, thereby completing the excavation cycle and preparing the machine for the next excavation operation.

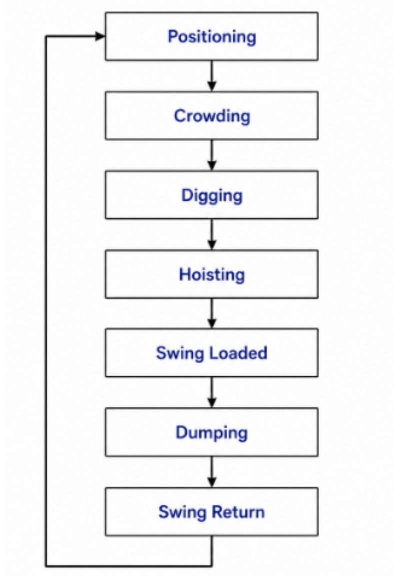


Figure 11. TTM

5.2 Seven-State Representation

To reproduce the operational behavior of the electric rope shovel, the excavation cycle is represented using seven distinct operating states. Each state corresponds to a specific phase of machine operation and is associated with unique mechanical loading conditions, payload characteristics, force distributions, and electrical drive requirements.

The adoption of a state-based representation enables the Digital Twin to capture the changing operating conditions encountered throughout the excavation cycle. Furthermore, state recognition provides a structured framework for coupling excavation mechanics with electrical drive system behavior.

The seven operational states considered in the proposed Timed Transition Model are described below.

5.2.1 Positioning

The positioning state represents the initial condition of the excavation cycle. During this phase, the bucket is located near the excavation face and the machine prepares for material engagement. Excavation forces are negligible, payload is absent, and the electrical drive system operates under relatively light loading conditions.

5.2.2 Crowding

In the crowding state, the crowd mechanism advances the dipper handle toward the material bank. The objective of this phase is to establish bucket penetration and position the bucket for excavation. Although excavation forces remain relatively small, crowd force gradually increases as the bucket approaches the material.

5.2.3 Digging

The digging state constitutes the primary excavation phase of the operating cycle. During this stage, the bucket penetrates the material bank and progressively accumulates payload. Excavation resistance reaches its highest levels during this phase due to cutting, lifting, and frictional interactions between the bucket and the material. Consequently, hoist force, crowd force, and motor load torque increase significantly.

5.2.4 Hoisting

Following completion of excavation, the hoisting state begins. The hoist mechanism lifts the loaded bucket away from the excavation face while supporting the combined weight of the bucket and excavated material. During this phase, payload-induced loading becomes the dominant contributor to motor load torque.

5.2.5 Swing Loaded

In the swing-loaded state, the upper structure rotates toward the dumping location while carrying the loaded bucket. Excavation resistance is no longer present; however, the electrical drive system continues to support the suspended payload during transportation.

5.2.6 Dumping

The dumping state represents the unloading phase of the excavation cycle. The bucket releases the excavated material at the designated dumping location, resulting in a progressive reduction in payload and suspended load. Consequently, motor loading decreases throughout this phase.

5.2.7 Swing Return

The swing-return state completes the excavation cycle. During this phase, the empty bucket is transported back toward the excavation face in preparation for the next excavation operation. Since the payload has been removed, loading conditions are substantially lower than those observed during the swing-loaded phase.

5.3 State Transition Logic

Timed Transition Model uses physically realistic criteria for making transitions between the states that are part of the digger's cycle. Each of these states continues to be operative until it meets the condition of transition to another state, after which the model moves to the next state.

Different from time-based approaches, the suggested model incorporates geometrical, payload-related, and other parameters to calculate transitions between states. The Digital Twin will thereby simulate the realistic process of digging that is consistent with the operation of the excavator.

Table 2. State transition logic employed in the Timed Transition Model

State Transition	Transition Condition	Physical Interpretation
Positioning → Crowding	$t \geq 4$ s	Bucket reaches excavation start position
Crowding → Digging	$d \geq 0.70$ m	Bucket penetrates the material bank to the required depth
Digging → Hoisting	$FF \geq 0.90$	Bucket reaches 90% fill factor
Hoisting → Swing Loaded	$y_b \geq 6.25$ m	Loaded bucket reaches the designated hoisting height
Swing Loaded → Dumping	$\theta_s \geq 90°$	Upper structure reaches the dumping location
Dumping → Swing Return	$FF \leq 0$	Bucket payload completely discharged
Swing Return → Positioning	$\theta_s \leq 0°$	Shovel returns to the excavation position

The transition conditions were selected to reflect the physical progression of the excavation cycle. Geometric variables such as excavation depth, bucket elevation, and swing angle are employed to represent machine motion, whereas fill factor is utilized to characterize payload accumulation and unloading. This combination of geometric and operational variables enables realistic state progression while maintaining computational simplicity.

The use of physically meaningful transition criteria also improves the interpretability of simulation results. Changes in force distribution, payload, motor torque, current demand, and power consumption can therefore be directly associated with specific operational states within the excavation cycle.

5.4 Timing Allocation of Operational States

Apart from considering the transition criteria, the realistic depiction of the excavation process involves the proper assignment of the operating times to each state as well. The time spent in each of the processes affects payload formation, force development, motor load, and efficiency of the whole system. Therefore, a realistic excavation cycle was formulated considering the operations of the rope shovel.

The complete excavation cycle considered in the present work has a duration of approximately 37 seconds. This cycle time is distributed among the seven operational states according to their relative significance within the excavation process.

Table 3. Timing allocation for the seven-state excavation cycle

Operational State	Duration (s)
Positioning	4
Crowding	1.5
Digging	10.3
Hoisting	5
Swing Loaded	7
Dumping	2
Swing Return	7.2
Total Cycle Time	**37**

The digging state occupies the largest portion of the cycle because excavation and bucket filling constitute the primary productive activity of the machine. During this phase, excavation resistance reaches its highest levels and the majority of payload accumulation occurs.

Hoisting and swing-loaded operations require additional time because the loaded bucket must be elevated and transported to the dumping location while supporting the excavated material. In contrast, positioning, crowding, and dumping are comparatively shorter operations.

The swing-return state is assigned a duration comparable to the swing-loaded state because both phases involve rotational motion of the upper structure. The resulting timing distribution provides a realistic representation of shovel operation while ensuring smooth progression through the excavation cycle.

5.5 Advantages of the Timed Transition Model

The Timed Transition Model is a systematic process by which the operation behavior of an electric rope shovel can be modeled. In doing so, the machine's operating behavior can be divided into states, and thus, mechanical and electrical properties can be assigned to different phases of the machine's operation.

One of the main benefits of the suggested method lies in its capability to account for the considerable variability of loading conditions encountered while excavating. The parameters such as resistance while digging, payload accumulation, hoist forces, crowd forces, motor torques, and energy consumed are not always the same within the working cycle but rather vary considerably depending on the particular operational mode. This allows the TTM to depict their variations in a physical sense.

Furthermore, the state-based approach also helps to increase interpretability of the model. Variations in the machine's actions may be attributed to clearly distinguishable operations like digging, lifting, swinging, and dumping.

The other strength of the TTM is the modular design, which allows for activation and customization of the mechanical, electrical, and control systems depending on the present operational status of the system, and does not affect the design framework as a whole. The modularity facilitates future growth of the Digital Twin framework and inclusion of more functionality into it.

In addition, the theory of TTM offers a perfect background for recognizing states of operation. This is because every state is defined by some stage of operation of the machine. The TTM theory can therefore be used in automated recognition of machine operations using sensory data.

Consequently, the Timed Transition Model serves as an effective mechanism for integrating excavation mechanics, electrical drive behavior, and Digital Twin functionality within a unified simulation environment.

6 Mechanical System Modeling

The mechanical subsystem of the Digital Twin is responsible for reproducing the excavation behavior of the electric rope shovel throughout the operating cycle. This subsystem describes bucket motion, excavation kinematics, digging resistance, payload accumulation, force transmission, and load generation mechanisms that occur during excavation.

Main purpose of using mechanical modeling approach is to calculate the forces on the bucket and to study the way how these forces act through the hoist and crowd mechanisms. Forces affect the electrical drive system via the load torque in the hoist motor.

In order to accomplish this goal, the geometry-based modeling approach has been used. The model consists of trajectory planning of bucket, estimation of excavation depth, computation of digging resistance, computation of fill factor, payload computation, force computation, and torque computation. This framework forms the mechanical basis of the Digital Twin and provides the required data for the electric drive subsystem.

6.1 Coordinate System and Geometric Representation

A two-dimensional Cartesian coordinate system is employed to describe the motion of the bucket during excavation. The excavation process is represented within the (x-y) plane, where the horizontal axis corresponds to the excavation direction and the vertical axis represents bucket elevation.

The bucket position is defined by the coordinates

[(x_b, y_b)]

which vary continuously throughout the excavation cycle. The boom point, saddle block, and other geometric reference locations are represented using fixed coordinates within the global reference frame.

The adopted coordinate system provides a convenient framework for describing bucket motion, excavation depth, rope orientation, crowd direction, and force transmission. Furthermore, the geometric representation enables the determination of bucket trajectory parameters required for digging resistance estimation and force decomposition calculations.

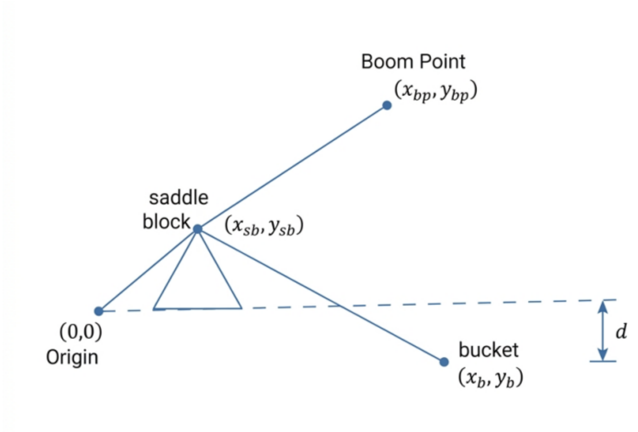


Figure 12. Coordinate system and geometric representation of the electric rope shovel

6.2 Bucket Trajectory Generation

The bucket trajectory governs the excavation path followed by the shovel during the digging process and forms the basis for excavation depth estimation, force generation, payload accumulation, and load transmission. In the present work, bucket motion is represented using a normalized excavation progress parameter (s), which varies continuously throughout the digging phase.

The normalized excavation progress is defined as

$$s = \frac{t_{state}}{t_{dig}} \tag{1}$$

where (t_{state}) denotes the elapsed time during excavation and (t_{dig}) represents the total digging duration. The parameter is bounded as

$$0 \leq s \leq 1 \tag{2}$$

such that (s=0) corresponds to the beginning of excavation and (s=1) corresponds to the completion of the digging operation.

The extension of the dipper handle is modeled using an exponential growth function:

$$L_h = 5.8 + 1.4\left(1 - e^{-2s}\right) \tag{3}$$

where (L_h) represents the instantaneous handle extension length. This formulation captures the rapid initial penetration of the bucket into the material bank followed by a gradual reduction in extension rate as excavation progresses.

The depth of cut is represented using a sinusoidal profile:

$$d = d_{max} \sin(\pi s) \tag{4}$$

where (d_{max}) denotes the maximum excavation depth.

Simultaneously, the bucket experiences a lifting motion during excavation. The lift profile is expressed as

$$y_{lift} = lift_{max} \sin(\pi s) \tag{5}$$

where ($lift_{max}$) represents the maximum bucket lift.

The bucket vertical position is therefore determined as

$$y_b = d + y_{lift} \tag{6}$$

where ($y_{surface}$) denotes the ground surface elevation.

The horizontal bucket position is obtained from the geometric relationship between the bucket and saddle block:

$$x_b = x_{sb} + \sqrt{L_h^2 - (y_b - y_{sb})^2} \tag{7}$$

where $((x_{sb}, y_{sb}))$ represents the saddle block coordinates.

The resulting trajectory provides a geometrically consistent representation of bucket motion and serves as the foundation for excavation kinematics, digging resistance estimation, and force decomposition calculations developed in the subsequent sections.

6.3 Excavation Kinematics

The excavation process involves the simultaneous motion of the bucket, hoist system, and crowd mechanism. Consequently, the determination of force directions is essential for accurately distributing excavation loads between the hoist and crowd subsystems.

To describe the excavation kinematics, three principal unit vectors are introduced: the digging direction vector, the hoist rope direction vector, and the crowd direction vector.

6.3.1 Digging Direction Vector

The digging direction vector defines the instantaneous direction of bucket motion through the material bank. It is obtained from the derivatives of the bucket trajectory with respect to the normalized excavation progress parameter (s).

The trajectory tangent vector is expressed as

$$\mathbf{t_d} = \begin{bmatrix} \frac{dx_b}{ds} \\ \frac{dy_b}{ds} \end{bmatrix} \tag{8}$$

The corresponding unit digging direction vector becomes

$$\mathbf{e_d} = \frac{\mathbf{t_d}}{|\mathbf{t_d}|} \tag{9}$$

where

$$|\mathbf{t_d}| = \sqrt{\left(\frac{dx_b}{ds}\right)^2 + \left(\frac{dy_b}{ds}\right)^2} \tag{10}$$

The vector (e_d) represents the direction along which excavation resistance acts during the digging process.

6.3.2 Hoist Rope Direction Vector

The hoist force acts along the rope connecting the bucket to the boom point. The rope direction vector is determined from the relative positions of the boom point and the bucket.

The rope position vector is given by

$$\mathbf{d_r} = \begin{bmatrix} x_{bp} - x_b \\ y_{bp} - y_b \end{bmatrix} \tag{11}$$

The corresponding unit rope direction vector becomes

$$\mathbf{e_r} = \frac{\mathbf{d_r}}{|\mathbf{d_r}|} \tag{12}$$

where

$$|\mathbf{d_r}| = \sqrt{(x_{bp} - x_b)^2 + (y_{bp} - y_b)^2} \tag{13}$$

The vector (e_r) defines the line of action of the hoist rope tension.

6.3.3 Crowd Direction Vector

The crowd force acts along the dipper handle connecting the saddle block to the bucket. The crowd direction vector is therefore obtained from the relative positions of these two points.

The crowd position vector is expressed as

$$\mathbf{d_c} = \begin{bmatrix} x_b - x_{sb} \\ y_b - y_{sb} \end{bmatrix} \tag{14}$$

The corresponding unit crowd direction vector becomes

$$\mathbf{e_c} = \frac{\mathbf{d_c}}{|\mathbf{d_c}|} \tag{15}$$

where

$$|\mathbf{d_c}| = \sqrt{(x_b - x_{sb})^2 + (y_b - y_{sb})^2} \tag{16}$$

The three unit vectors (e_d), (e_r), and (e_c) establish the geometric framework required for excavation force decomposition and load sharing calculations. These vectors enable the excavation resistance to be resolved into the corresponding hoist and crowd force components acting on the bucket.

6.4 Digging Resistance Modeling

The load offered by the resistance to the bucket during excavation is the key external load that acts on the shovel. The load caused by the resistance to excavation is due to the relationship between the bucket and the material bank..

Factors affecting the magnitude of the digging force include excavation depth, density of the excavated material, cohesion of the excavated material, and overburden force. With the increase in excavation depth, higher forces must be exerted by the bucket to overcome material resistance.

In the present work, the digging resistance is represented using a simplified physics-based formulation consisting of gravitational, cohesive, and surcharge components. The total digging resistance is expressed as

$$F_{dig} = \gamma g d^2 + c_{soil} d + q_{soil} \tag{17}$$

where

- γ is the material density,
- g is the acceleration due to gravity,
- d is the instantaneous depth of cut,
- c_{soil} is the equivalent soil cohesion parameter,
- q_{soil} is the surcharge resistance term.

The first term represents the contribution of material weight and increases quadratically with excavation depth. The second term accounts for the cohesive resistance of the material, while the final term represents a constant surcharge component associated with the excavation environment.

Using the parameters adopted in the simulation, the digging resistance becomes

$$F_{dig} = 1800 \times 9.81 \times d^2 + 22000, d + 9000 \tag{18}$$

This formulation produces increasing excavation resistance as the bucket penetrates deeper into the material bank, thereby reproducing the higher loading conditions typically encountered during the central portion of the digging phase.

6.4.1 Dynamic Excavation Disturbances

In practical excavation operations, the digging resistance is not perfectly smooth. Material heterogeneity, tooth impacts, rock fragmentation, and localized variations in material properties introduce fluctuations in the transmitted forces.

To capture these effects, disturbance components are superimposed on the nominal digging resistance. These disturbances represent phenomena such as tooth chatter, impact loading, rock breakage events, and transient force spikes that occur during excavation.

The resulting excavation load therefore consists of both the nominal digging resistance and the disturbance-induced force fluctuations, thereby providing a more realistic representation of excavation dynamics.

6.4.2 Effective Excavation Resistance

Not all of the generated digging resistance is transmitted directly to the shovel structure. The actual force transferred to the hoist and crowd subsystems depends on the excavation state and bucket engagement with the material.

To account for this behavior, an engagement factor is introduced. The effective excavation resistance is expressed as

$$F_{total} = e_f * F_{dig} \tag{19}$$

where (e_f) represents the excavation engagement factor.

The effective excavation resistance serves as the primary external load acting on the bucket and is subsequently resolved into hoist and crowd force components through the force decomposition procedure described in the following section.

6.5 Fill Factor Evolution and Payload Accumulation

The amount of material contained within the bucket during excavation is commonly represented using the fill factor. The fill factor provides a measure of bucket utilization and describes the fraction of the bucket volume occupied by excavated material.

During the digging phase, material progressively accumulates within the bucket as excavation proceeds. Consequently, the fill factor increases continuously from an initially empty condition to a nearly full bucket at the completion of excavation.

The fill factor is represented as

$$FF = \frac{V_{material}}{V_{bucket}} \tag{20}$$

where

- $V_{material}$ is the volume of material contained within the bucket,
- V_{bucket} is the rated bucket capacity.

To represent progressive bucket filling, the fill factor is modeled as an increasing function of excavation progress. During the digging phase, the fill factor evolves according to

$$FF(t + \Delta t) = FF(t) + e_{eng}\frac{\Delta t}{t_{dig}} \tag{21}$$

where

- e_{eng} is the excavation engagement factor,
- Δt is the simulation time step,
- t_{dig} is the digging duration.

The fill factor is constrained as

$$0 \leq FF \leq 1 \tag{22}$$

such that a value of zero corresponds to an empty bucket while a value of unity represents a completely filled bucket.

The transition from the digging state to the hoisting state occurs when the fill factor reaches a predefined threshold value

$$FF \geq FF_{crit} \tag{23}$$

where

$$FF_{crit} = 0.90 \tag{24}$$

indicating that the bucket has reached approximately ninety percent of its rated capacity.

6.5.1 Payload Accumulation

As the fill factor increases, the payload carried by the bucket also increases. The instantaneous payload weight is calculated as

$$W_{payload} = FF * W_{payload,max} \tag{25}$$

where

- $W_{payload}$ is the instantaneous payload weight,
- $W_{payload,max}$ is the maximum payload capacity.

The total bucket load acting during excavation is therefore given by

$$W_{total} = W_{bucket} + W_{payload} \tag{26}$$

where (W_{bucket}) represents the empty bucket weight.

Using the adopted model parameters,

$$W_{total} = 60000 + FF \times 25000 \tag{27}$$

The total bucket load increases continuously throughout the digging phase and reaches its maximum value at the completion of excavation. This increasing payload directly influences the hoist force, crowd force, and load torque acting on the electrical drive system.

6.6 Force Decomposition and Load Distribution

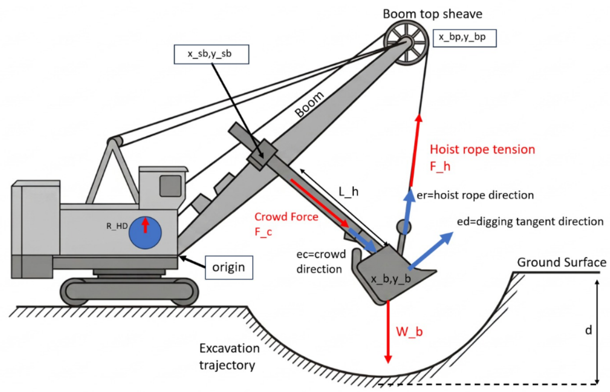
Once the excavation resistance and payload have been determined, the next step is to evaluate how these loads are distributed between the hoist and crowd mechanisms. During excavation, the bucket is simultaneously subjected to excavation resistance, gravitational loading, hoist rope tension, and crowd force. The interaction of these forces determines the loading transmitted to the shovel structure and ultimately to the electrical drive system.

To accurately represent this process, a geometry-based force equilibrium formulation is employed. Unlike simplified load-sharing approaches that assume fixed force distributions, the present method determines the load distribution directly from the instantaneous shovel geometry.

6.6.1 Bucket Force Equilibrium

The bucket is subjected to four principal forces:

1. Effective excavation resistance (F_{total})
2. Total bucket weight (W_{total})
3. Hoist force (F_H)
4. Crowd force (F_C)



The excavation resistance acts along the digging direction vector (e_d), while the hoist and crowd forces act along the rope direction vector (e_r) and crowd direction vector (e_c), respectively.

Applying static force equilibrium to the bucket yields

$$\mathbf{A}\begin{bmatrix} F_H \\ F_C \end{bmatrix} = \mathbf{RHS} \tag{28}$$

where

$$\mathbf{A} = \begin{bmatrix} e_{r,x} & e_{c,x} \\ e_{r,y} & e_{c,y} \end{bmatrix} \tag{29}$$

and

$$\mathbf{RHS} = \begin{bmatrix} F_{total}e_{d,x} \\ F_{total}e_{d,y} + W_{total} \end{bmatrix} \tag{30}$$

The matrix (A) contains the directional components of the hoist and crowd force vectors, while the vector (RHS) represents the external loads acting on the bucket.

6.6.2 Solution of the Force Equilibrium System

The hoist force and crowd force are obtained by solving the resulting system of linear equations:

$$\begin{bmatrix} F_H \\ F_C \end{bmatrix} = \mathbf{A}^{-1}\mathbf{RHS} \tag{31}$$

The resulting force components vary continuously throughout the excavation process as the bucket position, rope angle, excavation resistance, and payload change.

Consequently, the load distribution between the hoist and crowd mechanisms is not constant but evolves dynamically according to the instantaneous operating condition of the shovel.

6.6.3 Physical Interpretation

In the early stages of the excavation process, the crowd system plays an important role in bucket penetration. However, as excavation proceeds and the load increases, the role of the hoist system increases due to increasing bucket weight.

Towards the end of excavation, the effect of payload mass build-up and resistance to excavation causes maximum loading in the excavation cycle. This means that the load on the hoist system is very high and the resultant force is responsible for the torque on the hoist motor.

The computed forces of the hoist and the crowd act as the main output of the mechanical system and form the basis for the torque generated from the electrical drive system.

6.7 Load Torque Generation

The hoisting force obtained using the force equilibrium approach is the mechanical force that passes through the hoist rope system, hoist drum, gearbox, and the block and tackle system. In order to get the force exerted on the hoist motor, the above force should be translated to equivalent torque on the motor shaft. The hoist force acts on the hoist drum and generates a drum torque given by

$$T_{drum} = F_H r_{HD} \tag{32}$$

where

- F_H is the hoist force,
- r_{HD} is the hoist drum radius.

The torque transmitted to the motor shaft is reduced by the gearbox ratio and reeving arrangement. Consequently, the equivalent payload torque acting on the motor shaft is expressed as

$$T_{L,payload} = \frac{F_H r_{HD}}{n_{gr} n_{falls}} \tag{33}$$

where

- n_{gr} is the gearbox ratio,
- n_{falls} is the number of reeving falls.

This formulation establishes the direct relationship between bucket loading conditions and motor shaft loading.

6.7.1 Rope Dynamic Effects

The hoist cable, in real-world excavator systems, behaves in an elastic manner. This means that different loading conditions result in cable stretch and fluctuating tensions that affect the torque developed at the driveshaft.

The rope tension is represented as

$$F_{rope} = K_r x_r + C_r \dot{x}_r \tag{34}$$

where

- K_r is the rope stiffness,
- C_r is the rope damping coefficient,
- x_r is the rope stretch,
- $\dot{x}_r$ is the rope stretch velocity.

The corresponding rope-induced torque becomes

$$T_{L,rope} = \frac{F_{rope} r_{HD}}{n_{gr} n_{falls}} \tag{35}$$

This component captures transient loading effects arising from rope elasticity and dynamic force transmission.

6.7.2 Torque Ripple and Excavation Disturbances

Excavation operations are characterized by continuously varying loading conditions. Material heterogeneity, tooth impacts, rock fragmentation, and localized resistance changes introduce fluctuations in the transmitted torque.

To represent these effects, an additional torque ripple component is incorporated into the model:

$$T_{L,ripple} = f(t) \tag{36}$$

where (f(t)) represents the oscillatory torque fluctuations generated during excavation.

In addition, transient surge loads are introduced to capture sudden loading events such as tooth impacts and material breakage:

$$T_{surge} = g(t) \tag{37}$$

where (g(t)) represents short-duration torque spikes generated during excavation.

6.7.3 Total Load Torque

Load torque due to the action on the motor driving the hoist is evaluated by summation of payload torque, rope dynamics torque, torque ripple, and transient torque. Because the payload torque and rope-transferred torque result from the same force transfer route in quasi-static excavation mode, an averaging constant is included in order not to overestimate the load torque.

Thus, the total load torque becomes

$$T_L = \frac{1}{2}\left(T_{L,\text{rope}} + T_{L,\text{payload}}\right) + T_{L,\text{ripple}} + T_{\text{surge}} \tag{38}$$

The generated load torque is the main input to the electrical drive subsystem from a mechanical perspective. This coupling process makes it possible for any changes in resistance in the process of excavation, payload buildup, rope elasticity, and movement of the bucket to affect motor current, torque, angular velocity, and power consumption. Therefore, the load torque equation makes the coupling between the mechanical excavation model and the electrical drive model.

7 Electrical Drive System Modeling

The function of the drive subsystem involves the conversion of electrical power to mechanical torque to carry out the excavation process. While in operation, the motor that drives the hoist needs to adapt to changes that take place due to excavation resistance, load build up, rope dynamics, and states of transition during operation. Therefore, the electrical nature of the drive system becomes highly important.

In the well-developed Digital Twin concept, the electric part takes into account the load torque produced by the mechanical model and calculates the current of the motor, rotor flux, electromagnetic torque, speed of rotation, and energy consumption. In this way, changes in the excavating parameters will affect the electrical system directly.

The model of the electric drive comprises the sub-models of the rotor flux, torque production, shaft dynamics, speed controller, and current regulator. These elements work together to simulate the dynamic characteristics of the hoist drive system and build the electrical part of the Digital Twin.

7.1 Hoist Motor Specifications

The drive mechanism of an electric rope shovel comprises of a high-power induction motor whose main function is to provide the torque needed for digging, lifting, and loading. This induction motor forms the key element of energy conversion in the hoist drive mechanism and determines the efficiency of the machine.

The Digital Twin that has been built in the current research makes use of the rated parameters of the hoist motor in order to model the electrical subsystems. The parameters determine the operational constraints of the motor and also serve as reference values for the design and analysis of the control system.

Table 4 summarizes the principal motor specifications adopted in the simulation model.

Table 4. Hoist motor specifications used in the Digital Twin model

Parameter	**Value**
Rated Power	600 kW
Rated Voltage	690 V
Rated Current	628 A
Rated Frequency	25.4 Hz
Rated Speed	480 rpm
Rated Electromagnetic Torque	11937 N·m
Number of Pole Pairs	3
Rotor Resistance (R_r)	0.032 Ω
Rotor Inductance (L_r)	0.0445 H
Magnetizing Inductance (L_m)	0.042 H

Figure 13. Hoist motor

The rated speed and torque define the nominal operating condition of the hoist motor, while the electrical parameters are used in the rotor flux and torque generation equations. These specifications provide the foundation for the electrical drive model and establish the reference operating limits employed throughout the simulation.

The motor ratings also determine the allowable current limits, torque capability, and power consumption characteristics observed during the excavation cycle.

7.2 Crowd Motor Characteristics

The crowd mechanism pushes the dipper stick into the material face and furnishes the force necessary for digging operations. The motor has been constructed to deliver a high force at slow speeds in order to facilitate good engagement of the bucket with the material.

Table 5. Crowd Motor Specifications

Parameter	**Value**
Rated Power	350 kW
Rated Voltage	690 V
Rated Current	260 A
Rated Frequency	25.4 Hz

7.3 Swing Motor Characteristics

The swing drive causes the upper portion of the shovel to turn from the digging spot to the dump point. Swing motors are used to ensure smooth movement when the bucket is fully loaded.

Table 6. Swing Motor Specifications

Parameter	**Value**
Rated Power	260 kW
Rated Voltage	690 V
Rated Current	281 A
Rated Frequency	25.4 Hz

Figure 14. Swing Motor

7.4 Propel Motor Characteristics

The propel drive enables movement of the shovel within the mining environment. These motors provide the tractive effort required to reposition the machine between excavation locations.

Table 7. Propel Motor Specifications

Parameter	**Value**
Rated Power	450 kW
Rated Voltage	690 V
Rated Frequency	25.4 Hz

7.5 Rotor Flux Dynamics

Rotor flux is defined as the magnetic flux which is created inside the rotor circuit of the induction motor. Rotor flux can be considered to be one of the critical factors in generating electromagnetic torque in the machine.

In the present model, rotor flux dynamics are represented using a first-order differential equation derived from the rotor circuit equations of the induction motor.

The rotor flux evolution is expressed as

$$\frac{d\psi_{dr}}{dt} = -\frac{R_r}{L_r}\psi_{dr} + \frac{L_m}{L_r}R_r i_{ds} \tag{39}$$

where

- R_r is the rotor resistance,
- L_r is the rotor inductance,
- L_m is the magnetizing inductance,
- i_{ds} is the direct-axis current,
- ψ_{dr} is the rotor flux linkage.

The first term represents the natural decay of rotor flux due to rotor resistance, while the second term represents flux generation through the magnetizing current component.

The resulting rotor flux serves as the primary magnetic state variable of the motor and directly influences electromagnetic torque production.

7.6 Electromagnetic Torque Generation

EM torque arises due to the interaction between the magnetic field that is created inside the motor and the torque-inducing current in the stator. The resulting torque becomes the driving torque that helps overcome the loads during excavation and accelerate the motor shaft.

Within the adopted field-oriented control framework, the electromagnetic torque is expressed as

$$T_e = \frac{3}{2} p \frac{L_m}{L_r} \psi_{dr} i_{qs} \tag{40}$$

where

- p is the number of pole pairs,
- L_m is the magnetizing inductance,
- L_r is the rotor inductance,
- ψ_{dr} is the rotor flux linkage,
- i_{qs} is the quadrature-axis current.

The above expression shows that the torque is dependent on both rotor flux and the torque-producing component of current. As a result, the changes in current requirements due to excavation loading affect the electromagnetic torque of the motor.

The calculated electromagnetic torque acts as the primary driving torque within the motor shaft dynamics model.

7.7 Motor Shaft Dynamics

The relationship between the electrical drive system and the excavation system is controlled by the dynamics of the motor shaft rotation. The torque produced by the motor serves as the driving torque while the torque created by the load serves as the opposing torque to the rotation of the shaft. Moreover, there are viscous damping forces due to mechanical losses in the drivetrain.

Applying Newton's second law for rotational systems yields

$$J \frac{d\omega}{dt} = T_e + T_L - B\omega \tag{41}$$

where

- J is the equivalent rotational inertia,
- ω is the motor angular speed,
- T_e is the electromagnetic torque,
- T_L is the load torque,
- B is the viscous damping coefficient.

First of all, the first expression on the right side of the equation shows the torque caused by the action of the engine. Then, the second expression on the right-hand side is the torque that arises due to the resistance of loads and their transfer by means of the hoist equipment.

The relationship between the driving and resisting torques helps to determine the angular acceleration of the motor shaft. In case the electromagnetic torque is greater than the load torque, the motor accelerates. In case the load torque is greater than the electromagnetic torque, the motor decelerates.

The motor angular speed is obtained by integrating the shaft acceleration:

$$\omega(t) = \int \left(\frac{T_e - T_L - B\omega}{J} \right) dt \tag{42}$$

The resulting shaft speed provides direct insight into the interaction between excavation loads and motor capability. Consequently, variations in digging resistance, payload accumulation, and force transmission are reflected in the motor speed response throughout the excavation cycle.

Within the Digital Twin framework, the shaft dynamics subsystem serves as the principal coupling mechanism between the mechanical excavation model and the electrical drive model.

7.8 Speed Control System

Excavation operation is distinguished by the presence of load variation due to variations in resistance while digging, loading, and switching modes. Therefore, it is necessary to have a speed control system that maintains the optimum motor operation regardless of these disturbances.

For the fully developed Digital Twin, the feedback-based speed control algorithm has been used to control the speed of the motor. This controller keeps on comparing the reference speed with the actual motor speed and gives the command current for producing the torque.

The speed error is defined as

$$e_\omega = \omega_{ref} - \omega \tag{43}$$

where

- ω_{ref} is the reference motor speed,
- ω is the measured motor speed.

To eliminate steady-state error and improve dynamic performance, a proportional-integral (PI) controller is employed. The controller output is expressed as

$$u_{PI} = K_p e_\omega + K_i \int e_\omega dt \tag{44}$$

where

- K_p is the proportional gain,
- K_i is the integral gain.

Along with feedback control, feed-forward compensation is used to enhance the capability of disturbance rejection and fast response to changes in excavation loads. The feed-forward current is calculated based on the estimated torque of the load as follows:

$$i_{qs,ff} = \frac{T_L}{\frac{3}{2} p \frac{L_m}{L_r} \psi_{dr}} \tag{45}$$

The final quadrature-axis current reference is therefore calculated as

$$i_{qs,ref} = 0.72, i_{qs,ff} + K_p e_\omega + K_i \int e_\omega dt \tag{46}$$

The feed-forward component helps in predicting the changes in the excavation load condition, whereas the PI controller takes care of the remaining tracking error. Therefore, the drive mechanism can operate in a stable manner under varying digging load and payload conditions.

The generated current reference serves as the input to the current regulation subsystem, which determines the actual motor current and electromagnetic torque response.

7.9 Current Dynamics and Current Limiting

Induction motor torque is determined by the stator current vector components. Therefore, the correct description of current dynamics is required to simulate the realistic performance of the motor when digging.

In the developed model, both the direct-axis current and quadrature-axis current are represented using first-order dynamic systems. The direct-axis current evolves according to

$$i_{ds}(t + \Delta t) = i_{ds}(t) + \alpha_i \left(i_{ds,ref} - i_{ds}\right) \tag{47}$$

where

$$\alpha_i = \frac{\Delta t}{\tau_i} \tag{48}$$

and (τ_i) represents the current response time constant.

Similarly, the quadrature-axis current is calculated as

$$i_{qs}(t + \Delta t) = i_{qs}(t) + \alpha_i \left(i_{qs,ref} - i_{qs}\right) \tag{49}$$

The direct-axis current primarily controls rotor flux generation, whereas the quadrature-axis current determines electromagnetic torque production.

The overall stator current magnitude is obtained as

$$I = \sqrt{i_{ds}^2 + i_{qs}^2} \tag{50}$$

This quantity provides an indication of motor loading and is commonly used for performance assessment and condition monitoring purposes.

7.9.1 State-Dependent Current Limiting

During operation, the current demand varies significantly between different excavation states. Digging operations require large torque outputs to overcome excavation resistance, whereas positioning and return motions impose considerably lower loading requirements.

To account for these differences, state-dependent current limits are incorporated into the control framework.

During the digging state, the maximum allowable current is defined as

$$I_{max} = 1.35, I_{rated} \tag{51}$$

During the hoisting state,

$$I_{max} = 1.10, I_{rated} \tag{52}$$

For all remaining operational states,

$$I_{max} = 0.85, I_{rated} \tag{53}$$

This approach permits temporary current overload capability during demanding excavation operations while maintaining reduced electrical loading during less intensive operating states.

7.9.2 Current Saturation

To prevent excessive current commands and ensure stable operation, the quadrature-axis current reference is subjected to a smooth saturation function:

$$i_{qs,ref} = 0.985, I_{max} \tanh\left(\frac{i_{qs,ref}}{0.985, I_{max}}\right) \tag{54}$$

The hyperbolic tangent formulation provides a gradual transition into saturation and avoids abrupt controller nonlinearities that could otherwise introduce undesirable transient behavior.

The resulting current components determine the electromagnetic torque produced by the motor and therefore directly influence the mechanical response of the excavation system.

7.10 Motor Power Consumption

The energy consumption is one of the key performance parameters of the electric drive system. The changes in the excavation resistance, load accumulation and change in the operating conditions directly affect the amount of energy consumed during the excavation process.

The instantaneous mechanical power developed by the motor is obtained from the product of electromagnetic torque and motor angular speed:

$$P_m = T_e \omega \tag{55}$$

where

- P_m is the mechanical output power,
- T_e is the electromagnetic torque,
- ω is the motor angular speed.

For engineering analysis, power is commonly expressed in kilowatts. Consequently, the motor power is calculated as

$$P_m = \frac{T_e \omega}{1000} \tag{56}$$

where (P_m) is expressed in kW.

The momentary requirement of power changes during the excavation process based on the state of operation that is ongoing at that particular time. For instance, during the period when excavation is being done, high excavation resistance means that there will be an increase in torque requirements hence increased power usage. The same case occurs when hoisting is done.

In contrast, positioning, dumping, and swing-return operations generally impose lower mechanical loading conditions and therefore require comparatively lower power levels.

7.10.1 State-Dependent Power Characteristics

According to the Timed Transition model, the energy usage pattern can be linked to particular modes. In that case, typical energy signatures can be recognized for each stage of the digging process.

In the dig condition, the major factor contributing to power consumption is resistance and penetration. When in hoist condition, the load lifted is the most significant determinant of the power consumed. Swing loaded condition shows moderate power consumption due to transportation of the lifted load, and dump and swing return operations correspond to low loads.

The resulting power profile therefore provides valuable insight into the interaction between excavation mechanics and electrical drive performance.

7.10.2 Energy Consumption

The total energy consumed during an excavation cycle is obtained by integrating the instantaneous motor power over the simulation period:

$$E = \int P_m, dt \tag{57}$$

where (E) represents the cumulative energy consumption.

The calculated energy consumption provides a useful metric for evaluating machine efficiency, comparing operating scenarios, and assessing the influence of excavation conditions on overall drive system performance.

Consequently, motor power and energy consumption constitute key outputs of the Digital Twin framework and serve as important indicators of shovel operational behavior.

8 Simulation Framework and Model Integration

Developed Digital Twin includes various interconnected subsystems describing the process of excavating, electrical drive, state transition, and machine behavior. In isolation, each of the mentioned subsystems gives some information about a certain aspect of machine functioning. But in order to realistically model machine behavior, it is necessary to combine all these subsystems into one simulation.

The goal of the integration architecture is to determine the exchange of information among the Timed Transition Model, mechanical sub-system, and electrical drive sub-system. With the help of this coupling technique, excavation operations will affect the performance of the motor, whereas responses from the electrical system provide information about the performance of the equipment.

The resulting architecture enables the Digital Twin to reproduce the complete electromechanical behavior of the shovel throughout the excavation cycle.

8.1 Information Flow Architecture

Simulation takes place through an iterative exchange of information between different models. This process begins by determining the current operating state of the system through the Timed Transition Model for each simulation time period. This state then influences the operation of the mechanical and electrical systems.

The mechanical sub-system takes into account the following variables during excavation: bucket positioning, excavation depth, digging resistance, fill factor development, payload buildup, and force transmission. The above variables are employed to compute the load torque acting on the hoist motor.

The calculated load torque is transferred to the electrical subsystem, where rotor flux dynamics, current dynamics, electromagnetic torque generation, and shaft dynamics are evaluated. The resulting motor speed, current, torque, and power consumption are subsequently updated.

This sequential exchange of information continues throughout the simulation period, thereby enabling continuous interaction between the mechanical and electrical domains.

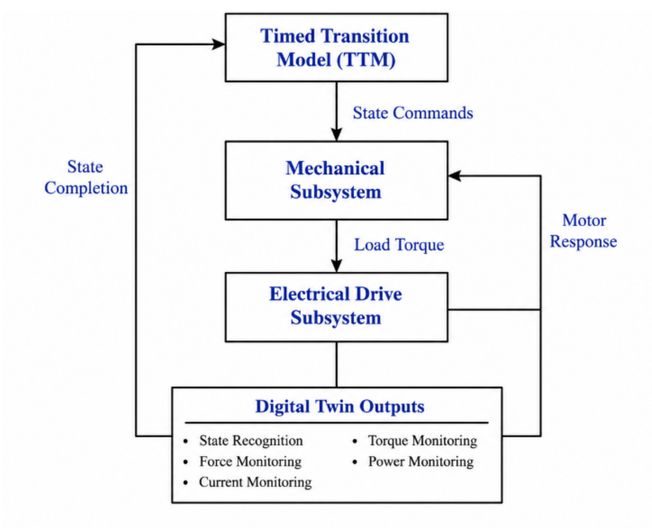


Figure 15. integration Architecture

8.2 Simulation Algorithm

Simulation of Digital Twin is done using discrete time computational approach. In each simulation time-step, state identification is first done using Timed Transition Model. The current state identifies the loading on the system, both mechanically and electrically.

The simulation procedure consists of the following sequence:

1. Determine the active operational state using the Timed Transition Model.
2. Update bucket trajectory and excavation geometry.
3. Compute excavation depth and digging resistance.
4. Evaluate fill factor evolution and payload accumulation.
5. Calculate hoist force and crowd force through force equilibrium.
6. Determine the equivalent load torque acting on the motor.
7. Update rotor flux dynamics.
8. Compute electromagnetic torque generation.
9. Solve motor shaft dynamics.
10. Update motor current, speed, torque, and power.
11. Store simulation results for post-processing.

This sequence is repeated until completion of the excavation cycle.

8.3 Model Coupling Strategy

The connection between the two subsystems is made using the torque due to the load from the digging operation. The mechanical model calculates the forces that act on the bucket and the hoist force that acts through the rope system as a consequence.

The hoist force is thereafter converted to equivalent torque of motor by gearbox and reeving system model. This motor torque becomes the key input to the electric drive system.

In the context of the electrical model, the effect of the load torque affects rotor current requirements, electromagnetic torque production, speed of the motor, and power consumption. Therefore, any changes in excavation resistance, load collection, and transition of operating states manifest themselves through the electrical model of the drive system.

This coupling strategy establishes a bidirectional relationship between excavation mechanics and electrical performance, thereby enabling realistic simulation of the complete shovel system.

8.4 Digital Twin Outputs

The integrated simulation framework generates a range of mechanical and electrical outputs that collectively describe the operating condition of the shovel.

Mechanical outputs include:

- Bucket position
- Excavation depth
- Digging resistance
- Fill factor
- Payload weight
- Hoist force
- Crowd force
- Load torque

Electrical outputs include:

- Rotor flux

- Motor current
- Electromagnetic torque
- Angular speed
- Mechanical power
- Energy consumption

Together, these outputs provide a comprehensive representation of shovel behavior and form the basis for performance analysis and future Digital Twin enhancements.

9 Results and Discussion

The Digital Twin model that was developed was used for simulating the entire process of excavation performed by the electric rope shovel. The simulation includes the Timed Transition Model, the mechanical excavation system, and the electrical drive system.

The aim of the simulation model is to study the interactions between the excavation processes and the efficiency of the electric drive system during the entire process of operation. There is a specific focus on changes in resistance to excavation, payload, load transfer, load torque, electric current of the motor, torque and energy consumption.

Simulation results from this section help in understanding the operating behavior of the shovel and show the ability of the suggested Digital Twin architecture in replicating the real-life behavior of the machine.

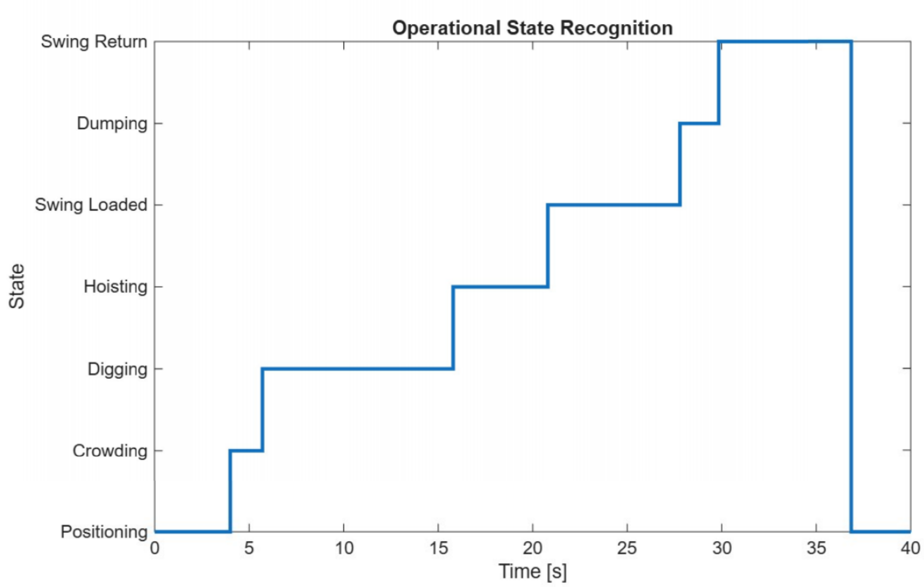


Figure 16. Evolution of operational states throughout the excavation cycle.

The sequence of transitions is able to illustrate that the Timed Transition Model can simulate realistic excavation processes while giving state-based loads on the mechanical and electrical subsystems.

The longest period corresponds to the process of digging because it involves the main work process of the excavator, which consists of digging and loading. On the other hand, the processes of positioning and unloading take less time than that.

9.1 Depth of Cut Variation

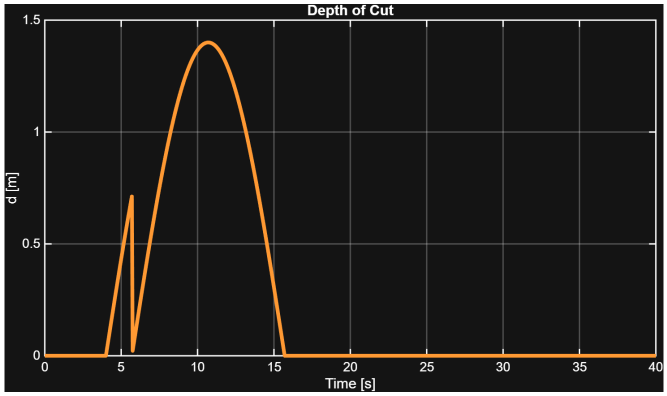


Figure 17. Variation of excavation depth during the excavation cycle.

Figure 17 shows the development of cut depth during the entire excavation operation. Cut depth varies according to the planned excavation path and grows continuously as the bucket moves into the material bank.

This is where the maximum depth is attained at about half way through the digging process. This corresponds to the maximum penetration stage. After this point, the depth of excavation becomes less with the approach of the end of the digging process.

The gradual variation in excavation depth is an indication of the efficiency of the selected trajectory generation approach and serves as the geometrical basis of digging resistance determination. Due to the fact that digging resistance is directly related to depth of cut, the profile obtained has a great influence on force transmission and motor loading.

9.2 Load Torque Evolution

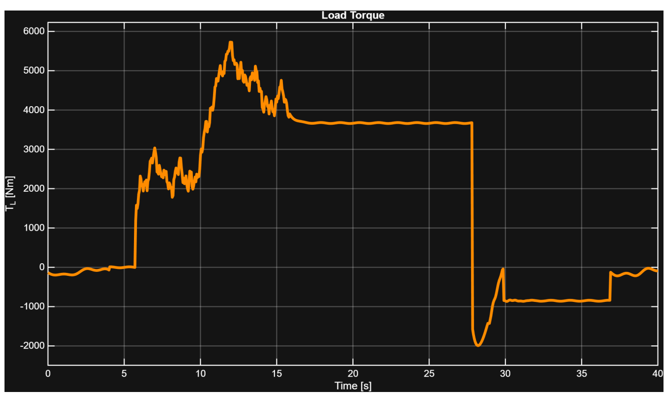


Figure 18. Load Torque

Figure 18 illustrates the total load torque acting on the hoist motor throughout the excavation cycle. The load torque represents the combined effect of payload loading, rope dynamics, excavation disturbances, and transient surge events.

In positioning and crowding, the load torque will always be small since there is little excavation resistance and payload loading. Once the bucket starts moving into the digging phase, the excavation resistance becomes large, thus increasing the motor load.

The highest load torque values occur near the completion of excavation when both excavation resistance and payload accumulation simultaneously reach elevated levels. Under these conditions, the hoist subsystem must support the combined influence of bucket weight, payload weight, and excavation forces.

The oscillations that have been noted in the behavior of the load torque can be attributed to torque ripple, which has been used to model the excavation disturbances including tooth impact, material breakage, and local differences in material resistance.

The resulting torque profile demonstrates the complex and highly dynamic nature of excavation loading and highlights the importance of incorporating realistic disturbance mechanisms within the Digital Twin framework.

9.3 Electromagnetic Torque Evolution

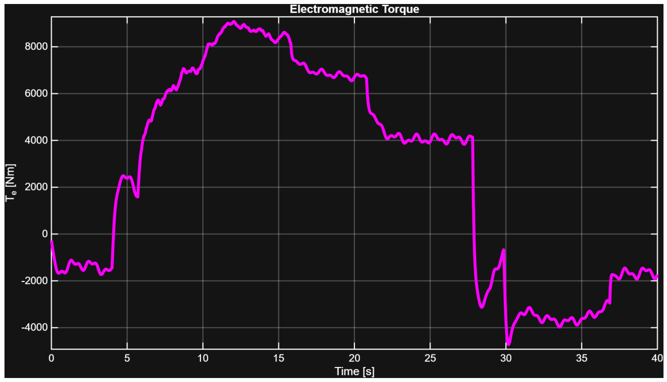


Figure 19. Electromagnetic torque

The electromagnetic torque of the hoist motor during the excavation process is depicted in Figure 19. This torque depends on the relationship between the rotor flux and the quadrature-axis current.

The generated electromagnetic torque closely follows variations in load torque throughout the operating cycle. As excavation resistance and payload loading increase, the control system demands additional torque-producing current, resulting in higher electromagnetic torque output.

While digging and lifting, the motor creates significantly higher torque because of the loads due to excavation and holding up the load itself. On the other hand, when positioning and dumping, there is less load, which causes lesser torque requirements.

The close correspondence between the load torque and electromagnetic torque responses demonstrates the effectiveness of the speed control system in maintaining stable machine operation under varying excavation conditions.

Small oscillations visible in the response arise from the disturbance components incorporated into the model and represent realistic transient loading conditions encountered during excavation operations.

9.4 Motor Speed Evolution

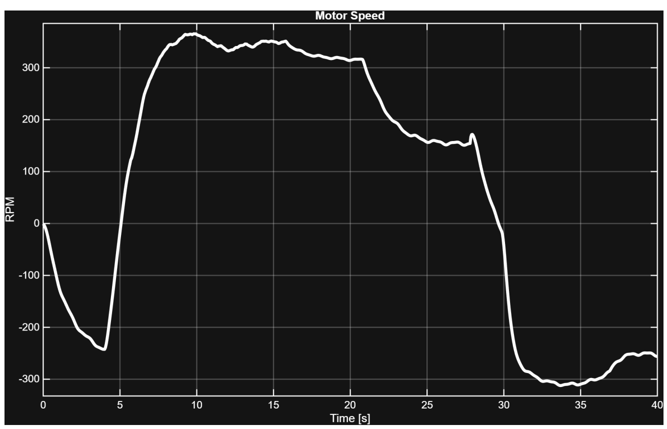


Figure 20. Motor Speed

Figure 20 illustrates the motor speed response throughout the excavation cycle. The speed profile reflects the interaction between the reference speed command, load torque variations, and controller action.

Different speed rates have been observed depending on the mode of operation. Faster speeds are usually characteristic of digging whereas slower speed is characteristic of swinging, positioning and dumping modes. This is in line with the state-specific speed references used in the Timed Transition Model.

Despite substantial changes in excavation loading conditions, the motor speed remains stable and closely follows the commanded operating profile. This behavior demonstrates the capability of the control system to compensate for load disturbances and maintain satisfactory speed regulation.

The transient behavior that occurs during transition from one state to another is a result of fast variation in the load torque. This is an expected behavior of an actual excavator system and gives us information about the interaction between the two subsystems.

9.5 Motor current Evolution

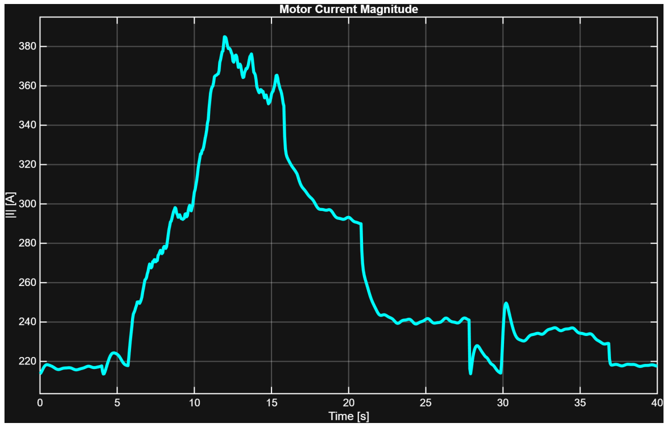


Figure 21. Motor Current

Figure 21 shows the variation of the magnitude of motor current during the excavating cycle. The reason for this is that the generation of electromagnetic torque is directly proportional to the amount of current required.

The magnitude of current rises sharply in the digging cycle owing to the need for more torque to overcome the resistance offered by the digging process and due to the loading. An increased magnitude of current will also be experienced during hoisting because of the load.

The state-dependent current limiting strategy effectively constrains motor loading while still providing sufficient torque capability during demanding excavation operations. Consequently, the current response remains within acceptable operating limits throughout the simulation.

The observed current variations demonstrate the strong relationship between excavation mechanics and electrical drive performance, thereby validating the electromechanical coupling implemented within the Digital Twin framework.

9.6 Motor power Evolution

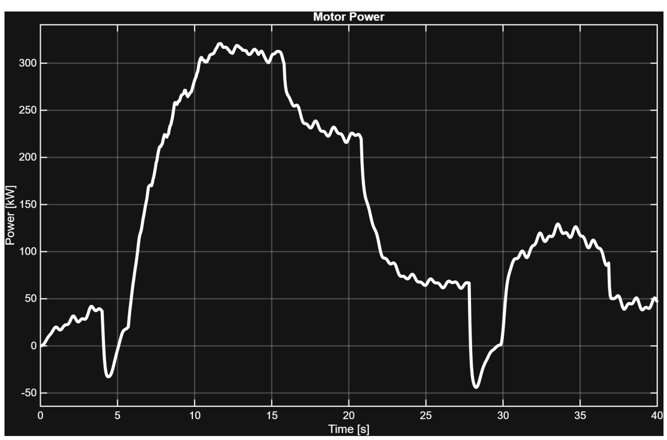


Figure 22. Motor Power

Figure 22 illustrates the change in power consumption of the motor during the entire excavation process. The power consumption depends on the effect of both the production of electromagnetic torque and the speed of the motor rotation.

The requirement of power varies according to the operation status. At the position and return states, there is low load, and therefore power requirement is low. When the bucket starts digging, the resistance to digging increases and thus there will be high torque required from the motor and thus power requirement is high.

Maximum power is usually realized during digging and hoisting cycles. Digging entails the need for overcoming the resistance from the excavation process as well as lifting loads. Hoisting involves the power needed for raising the loaded bucket from the ground.

The power variations that are encountered during the response process result from loading variations, torque ripple effect, and changes in excavation resistance. The variations are typical of practical excavation systems, and they reflect the dynamics of shovel operations.

The anticipated power characteristic emphasizes the high level of correlation between excavation dynamics and the dynamics of electric drive operation. For this reason, power consumption becomes a key parameter of the machine's performance and an invaluable source of data for productivity evaluation and energy management purposes.

9.7 Discussion of Electro-Mechanical Coupling

The simulation results show the strong coupling effect between excavation mechanics and the behavior of the electrical drive system throughout the excavation process. The position and depth of the bucket play a vital role in determining the resistance to be faced during the excavation operation. This resistance is later divided into the hoist and crowd systems via force decomposition using the geometric model.

The force determined on the hoist is converted to the load torque for the motor using the hoist drum, gearbox, and reeving system model. Therefore, any changes in resistance to excavation and payload loading will be directly translated into torque requirements by the motor system.

As the load torque is increased, the speed control mechanism acts to increase the current producing torque, resulting in more electromagnetic torque. Through this process, the machine is able to operate at the required point regardless of the magnitude of loading encountered during the excavation process.

The obtained results on the speed, current, torque, and power consumption show that there is a successful integration of the mechanical and electric systems into the digital twin. Therefore, the model created offers a realistic representation of the shovel operations and will be the basis for conducting future researches on condition monitoring, predictive maintenance, performance optimization, and advanced control.

10 Conclusion

The paper has covered the creation of a multiphysics digital twin of an electric rope shovel for use in diagnosis and prognosis of faults. The framework created is capable of incorporating excavation mechanics, force breakdown, load creation, and recognition of the machine's operational status into one model.

A geometry-based kinematic model was developed for generating realistic bucket path during excavation. This model incorporates the interrelation between crowding and lifting actions while preserving geometric consistency between the bucket, dipper stick, saddle block, and boom assembly. The resultant bucket path was then employed for determining the depth of excavation, contact of material, and digging force direction.

To represent bucket-material interaction, a simplified McKyes-inspired digging resistance model was implemented. The formulation incorporates the effects of material density, cohesion, surcharge loading, and excavation depth while maintaining computational efficiency suitable for repeated digital twin simulations. Additional disturbance components were introduced to represent tooth chatter, tooth impacts, rock breakage, and sudden excavation resistance variations.

The vector force decomposition methodology was introduced in order to allocate loads for excavation to the hoist and crowd system depending on the instant geometry of the shovel. This method allows for the realistic estimation of hoist and crowd forces without using the preconceived load distribution ratios. Hoist force obtained above was converted to the motor load torque based on reeving, hoist drum and gearbox.

The results obtained through simulation have proved that the designed model is able to replicate the behavior of excavation including the trajectory path of the bucket, change in force while excavating, load distribution between the hoist and crowd system, and the torque in the loads as a result of disturbance during the excavation process.

Overall, the developed digital twin can serve as an efficient computational framework for analysis of shovel's performance and serves as a basis for further development of fault diagnosis and prognostic tools.

References

[1] T. D. Lopes, A. Raizer, and W. Valente Jr., "The use of digital twins in finite element for the study of induction motors faults," *Sensors*, vol. 21, no. 23, p. 7833, 2021.

[2] R. Gulati *et al.*, "MINDTwin AI: Multiphysics informed digital-twin for fault localization in induction motor using AI," in *Proc. IEEE IECON*, 2023, pp. 1–6.

[3] Y. Liu *et al.*, "Numerical simulation and experiment on excavating resistance of an electric cable shovel based on EDEM-RecurDyn bidirectional coupling," *Machines*, vol. 10, no. 12, p. 1203, 2022.

[4] S. I. Malafeev and V. I. Konyashin, "Induction motor drives for electric mining shovels: Synthesis, design and research," in *Proc. IEEE ICIEAM*, 2018, pp. 1–5.